

Investigate how advances in technologies enable advances in science, and how science has contributed to developments in technologies and engineering

Learning intention

- investigate how advances in technologies enable advances in science, and how science has contributed to developments in technologies and engineering.

Teach and model

- exploring how scientists and engineers make machines more energy efficient.
- exploring how understanding of the nature of matter and energy has changed over time.
- examining how advances in understanding of radioactivity and radioisotopes have led to new applications and technologies.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.